

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second Semester of M. Sc. (Mathematics) Examination April-May 2019
MA722 Differential Geometry

Date : 29/04/2019, Monday

Time: 10:00 a.m. to 10:45 a.m.

Total marks 20

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Maximum time to answer Multiple Choice Questions will be 45 minutes.
4. Each question carries one mark.
5. All notations and terminologies are standard.

Q-I Choose the correct answer from the given options in the followings:

[20]

- (1) Which of the following statement is not true in curve theory?
 - (a) A reparametrization of a regular curve is regular.
 - (b) The absolute value of the signed curvature is the curvature of the curve.
 - (c) Any simple closed curve in $\mathbb{R}^2$ has an “interior” and “exterior”.
 - (d) In Wirtinger’s inequality, the equality holds if and only if there is a constant A such that $F(t) = A \cos t$ for all t .
- (2) In isoperimetric inequality, if $\bar{\gamma}$ is a circle then $[l(\bar{\gamma})]^2 =$ _____.
 - (a) 4π
 - (b) $A(\bar{\gamma})$
 - (c) $\pi A(\bar{\gamma})$
 - (d) $4\pi A(\bar{\gamma})$

P.T.O.

- (3) Which of the following is not a vertex of the curve $\bar{\gamma}(t) = (a \cos t, b \sin t)$?
- (a) $\bar{\gamma}(0) = (a, 0)$ (b) $\bar{\gamma}\left(\frac{\pi}{2}\right) = (0, b)$
(c) $\bar{\gamma}(\pi) = (0, -a)$ (d) $\bar{\gamma}\left(\frac{3\pi}{2}\right) = (0, -a)$
- (4) The curve $\bar{\gamma}(t) = (a \cosh t, b \sinh t)$ represents _____.
(a) line (b) parabola
(c) hyperbola (d) circle
- (5) The value of $R_\theta(x, y)$ for $\theta = \frac{\pi}{4}$ is _____, where $R_\theta(x, y)$ is rotation operator.
- (a) $\left(\frac{x-y}{\sqrt{2}}, \frac{x+y}{\sqrt{2}}\right)$ (b) $\left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right)$
(c) $\left(\frac{xy}{\sqrt{2}}, \frac{xy}{\sqrt{2}}\right)$ (d) $\left(\frac{\sqrt{2}}{x+y}, \frac{\sqrt{2}}{x-y}\right)$
- (6) The equation of the tangent space to the unit sphere at $(0,1,0)$ is _____.
(a) $y = 1$ (b) $y = 0$
(c) $x = 1$ (d) $x + z = 0$
- (7) A Cartesian representation of the curve $\bar{\gamma}(t) = (\cos^3 t, \sin^3 t)$ is _____.
(a) $x + y = 1$ (b) $x^2 + y^2 = 1$
(c) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$ (d) $x^{\frac{3}{2}} + y^{\frac{3}{2}} = 1$
- (8) Which of the following surfaces can be covered with one patch only?
(a) Ellipsoid (b) Sphere
(c) Hyperboloid with two sheets (d) Torus
- (9) The first fundamental form on the surface $\sigma(u, v) = (u, v, u - v)$ is _____.
(a) $du^2 + dv^2$ (b) $du^2 - dv^2$
(c) du^2 (d) *None of these*
- (10) Which of the following is not a surface?
(a) $\mathcal{S} = \{(x, y, z) \in \mathbb{R}^3 : z = y^2 - x^2\}$
(b) $\mathcal{S} = \{(x, y, z) \in \mathbb{R}^3 : z^2 = x^2 + y^2, z \geq 0\}$
(c) $\mathcal{S} = \{(x, y, z) \in \mathbb{R}^3 : z^2 = x^2 + y^2\}$
(d) *None of these*
- (11) Which of the following statement is not true?
(a) Every meridian is a geodesic.
(b) Any normal section on a surface is a geodesic.
(c) A unit speed curve $\bar{\gamma}$ is geodesic if and only if its geodesic curvature is zero.
(d) The image of the Gauss map is a circle for a plane.
- (12) Which of the following surface has the constant Gaussian curvature?
(a) sphere (b) hyperboloid
(c) torus (d) none of these
- (13) The point $(0,0,0)$ on the surface $z = x^2 + y^2$ is _____.
(a) parabolic (b) elliptical
(c) hyperbolic (d) flat
- (14) The principal curvatures of a surface $z = y^2 - x^2$ are _____.
(a) $2, -2$ (b) $0, 2$
(c) $2, 2$ (d) $1, 1$

P.T.O.

- (15) The Gaussian curvature and mean curvature of a surface does not exist if _____.
 (a) $EG - F^2 = 0$ (b) $LN - M^2 = 0$
 (c) $EG - F^2 \neq 0$ (d) $LN - M^2 \neq 0$
- (16) Which of the following theorem gives intrinsic property of the surface?
 (a) The Gauss-Bonnet theorem (b) The Gauss's remarkable theorem
 (c) Bonnet theorem (d) Euler's theorem
- (17) The sum of interior angles of a triangle on a sphere is _____.
 (a) greater then π (b) equal to π
 (c) greater and equal to π (d) equal to 2π
- (18) The geodesic curvature of the plane is _____.
 (a) 0 (b) 2π
 (c) π (d) 1
- (19) Let $\sigma: U \rightarrow \mathbb{R}^3$ be a surface patch. Which of the following statements is correct?
 (a) $(\sigma_{uu})_v = (\sigma_{vu})_v$ (b) $(\sigma_{uu})_u = (\sigma_{vu})_v$
 (c) $(\sigma_{uu})_v = (\sigma_{uv})_u$ (d) $(\sigma_{uv})_v = (\sigma_{vu})_u$
- (20) Let $\sigma: U \rightarrow \mathbb{R}^3$ be a surface patch. Which of the following statements is not true?
 (a) $F\mathcal{K} = \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 + (\Gamma_{12}^1)_u - (\Gamma_{11}^1)_v$
 (b) $F\mathcal{K} = \Gamma_{12}^2 \Gamma_{12}^1 + \Gamma_{11}^2 \Gamma_{22}^1 - (\Gamma_{12}^1)_u + (\Gamma_{11}^1)_v$
 (c) $F\mathcal{K} = \Gamma_{12}^1 \Gamma_{12}^2 - \Gamma_{22}^1 \Gamma_{11}^2 + (\Gamma_{12}^2)_v - (\Gamma_{22}^2)_v$
 (d) $F\mathcal{K} = \Gamma_{12}^1 \Gamma_{12}^2 - (\Gamma_{11}^1)_v + (\Gamma_{12}^1)_u - \Gamma_{22}^1 \Gamma_{11}^2$

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
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MA722 Differential Geometry

Date: 29/04/2019, Monday Time 10:45 a.m. to 1:00 p.m. Total marks of Section I & II 50

Instructions:

1. Section I and Section II must be written in separate answer books.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations and terminologies are standard.

SECTION - I

Q-II Answer the followings:

(A) Attempt any *TWO* of the following:

[8]

- (1) Prove that a point p on a surface S is umbilic if and only if Weingarten map is a scalar multiple of the identity map.
- (2) (i) Let p be a point on a surface S . Prove that the mean curvature $\mathcal{H}$ and the Gaussian curvature $\mathcal{K}$ at p are given by $\mathcal{H} = \frac{1}{2}(\mathcal{K}_1 + \mathcal{K}_2)$ and $\mathcal{K} = \mathcal{K}_1 \cdot \mathcal{K}_2$ where $\mathcal{K}_1$ and $\mathcal{K}_2$ be principal curvatures of S .
(ii) Show that the sum of interior angles of a regular n -gon on a plane is $(n - 2)\pi$.
- (3) (i) Show that Weingarten map at a point p is a self adjoint map.
(ii) State Bonnet's theorem and Gauss's remarkable theorem.

(B) Attempt any *SIX* of the following:

[12]

- (1) Compute the signed unit normal and signed curvature of the curve

$$\bar{\gamma}(t) = \left(R \cos \left(\frac{t}{R} \right), R \sin \left(\frac{t}{R} \right) \right).$$

- (2) Define Frenet- Serret equations.
- (3) Show that $R_\theta(x, y)$ is an isometry.
- (4) Let $\bar{\gamma}$ be a parameterized curve. If $\|\bar{\gamma}(t)\| = 1$ for all t , then show that $\bar{\gamma}(t) \perp \dot{\bar{\gamma}}(t)$, for all t .
- (5) Evaluate $EG - F^2$ of the surface patch $\sigma(u, v) = (u, v, u^2 - v^2)$, where E, F and G are the first fundamental magnitudes.
- (6) Let $\sigma: U \rightarrow \mathbb{R}^3$ be a surface patch. Show that $\|\sigma_u \times \sigma_v\| = \sqrt{EG - F^2}$.
- (7) Define area of the interior of the curve and conformal map of the surface.
- (8) Show that any geodesic has a constant speed.
- (9) Let σ be a surface patch of S and $p \in S$. Prove that $\mathcal{H} = \frac{LG - 2MF + NE}{2(EG - F^2)}$.

P.T.O.

SECTION – II

Q-III Answer the followings:

(A) Attempt any *THREE* of the following:

[18]

- (1) Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be smooth surfaces and $f: \mathcal{S}_1 \rightarrow \mathcal{S}_2$ be a smooth map. Show that f is a local isometry if and only if the symmetric bilinear maps $\langle \cdot, \cdot \rangle_p$ and $f^* \langle \cdot, \cdot \rangle_p$ are equal for every $p \in \mathcal{S}_1$.
- (2) Let $\sigma: U \rightarrow \mathbb{R}^3$ be a surface patch. Show that Christoffel symbols are defined as

$$\Gamma_{11}^1 = \frac{GE_u - 2FF_u + FE_v}{2(EG - F^2)}$$

and

$$\Gamma_{12}^1 = \frac{GE_v - FG_u}{2(EG - F^2)}.$$

- (3) Compute the Gaussian and Mean curvature of the unit cylinder.
- (4) State and derive Codazzi-Mainardi's equations.
- (5) State and derive Gauss's equations.

(B) Attempt any *TWO* of the following:

[12]

- (1) (i) Let $\mathcal{K}: (a, b) \rightarrow \mathbb{R}$ is a smooth map. Prove that there is a unit speed curve $\bar{\gamma}: (a, b) \rightarrow \mathbb{R}^2$ whose signed curvature is $\mathcal{K}$.
- (ii) Let σ be a surface patch of an oriented surface $\mathcal{S}$ containing a point p of $\mathcal{S}$, say, $p = \sigma(u, v)$. Show that $II\langle w, x \rangle_p = Ldu(w)du(x) + M[du(w)dv(x) + du(x)dv(w)] + Ndv(w)dv(x)$. For all $w, x \in T_p\mathcal{S}$, where $du, dv: T_p\mathcal{S} \rightarrow \mathbb{R}$ are defined by $du(w) = \lambda$ and $dv(w) = \mu$ when $w = \lambda\sigma_u + \mu\sigma_v \in T_p\mathcal{S}$.
- (2) Let $\bar{\gamma}: (a, b) \rightarrow \mathbb{R}^3$ be a regular curve with nowhere vanishing curvature. Prove that the torsion τ of $\bar{\gamma}$ is given by the formula

$$\tau = \frac{(\dot{\bar{\gamma}} \times \ddot{\bar{\gamma}}) \cdot \dddot{\bar{\gamma}}}{\|\dot{\bar{\gamma}} \times \ddot{\bar{\gamma}}\|^2}.$$

- (3) (i) If $f: \mathcal{S}_1 \rightarrow \mathcal{S}_2$ and $g: \mathcal{S}_2 \rightarrow \mathcal{S}_3$ are smooth maps, prove that

$$D_p(g \circ f) = D_{f(p)}g \circ D_p f, \text{ for all } p \in \mathcal{S}_1.$$

- (ii) Let p be a point on an oriented surface $\mathcal{S}$, and $\mathcal{K}_1$ and $\mathcal{K}_2$ be principal curvatures of $\mathcal{S}$ with principle directions $\bar{t}_1$ and $\bar{t}_2$. Let v be a unit vector in $T_p\mathcal{S}$. Show that the normal curvature of $\mathcal{S}$ at p in the unit direction v is given by $\mathcal{K}_n = \mathcal{K}_1 \cos^2 \theta + \mathcal{K}_2 \sin^2 \theta$ where θ is the oriented angle $\widehat{\bar{t}_1 v}$.

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